DA-IICT Gandhinagar

EL 111-- Basic Electronic Circuits (3-0-2-4)

AY: 2025-2026

Course Placement: Basic Electronic Circuits is a core UG course for 1st year (1st semester) B.Tech ICT program.

Course Format: EL 111-- **3 hours' lecture** and **2 hours' lab** every week.

Course Objective:

The objective of this course is to provide students the fundamental idea of electronic circuits and components. The course will enable students with the skills of working with different circuit elements like resistor, capacitor, inductor, diode, transistor, operational amplifier. The content is designed to help students not only to analyze a circuit but also to design an electronic circuit with necessary components.

Course Content:

1. Need for Electrical and Electronic Circuits, Applications of Electronic circuits, LT Spice based analysis of circuits
2. Signals and Circuits and Systems: General ideas
3. Voltage and Current Sources, Transducers, Batteries, Power Supplies. Passive and Active Components. Circuit Laws: KCL, KVL. Mesh and Nodal Analysis.
4. Network Theorems: Thevenin and Norton's Theorems, Superposition, Maximum power transfer, Source Transformation.
5. Ideal OP AMP. OP AMP Circuits with Resistances.
6. Ideal Diodes and Diode-Resistance Circuits
7. Energy Storage Elements – Inductors, Capacitors
8. Basics of RL and RC circuits: Source-free and driven, Natural and Forced Response
9. Sinusoidal Steady-State Analysis: Characteristics of Sinusoids, Forced Response to Sinusoidal Function, Phasors, Impedance, Circuit Laws, Network Theorems
10. Filters: The concept of LPF, HPF, BPF, and BSF, Frequency Response, Cut-off Frequency, Bandwidth, Quality Factor Ideas. Designing Passive and Active Filters.
11. BJT: Operating region, Biasing circuits, Small signal operation, Amplifier design.

Readings:

1. Lecture Notes
2. R. Dorf and J. Sbovoda, *Introduction to Electrical Circuits,* 6th Ed., John Wiley, 2006.
3. Charles K Alexander and Matthew N. O. Sadiku, *Fundamentals of Electric Circuits*, 4th Ed., McGraw Hill, 2009.
4. Adel S. Sedra and Kenneth C. Smith, *Micro Electronic Circuits: Theory and Applications*, 5th Ed., Oxford University Press, 2009.
5. Jacob Millman and Christos Halkias, *Integrated Electronics: Analog and Digital Circuits and Systems*, 2nd Ed., McGraw Hill, 2017.
6. J. O'Malley, *Schaum's Outline of Basic Circuit Analysis* 2nd. Ed., McGraw Hill, 1992.

Assessment Method (Tentative):

- 1st In-semester examination – 15%
- 2nd In-semester examination -- 20%
- End-semester examination – 40%
- Lab Experiments, Reports, and Lab Exam – 20%
- Class interactions – 5%

Course Outcomes:

After completion of this course, students should be able to:

- Understand the working principles of electronic circuits. [P1, P12]
- Analyze a given circuit using mathematical knowledge. [P1, P2, P4]
- Identify any issues in a circuit, and propose a solution. [P2, P3, P4]
- Work in a group for a laboratory experiment, and present their results [P9, P10].

Mapping of Course Outcomes to Program Outcomes:

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
x	x	x	x					x	x		x

Lecture Schedule:

Sl. No.	Description		No. of Lectures
1	**Introduction to the Course and Lab**		**2**
	1.1	Basic Concept of Electronic Circuits	
	1.2	Active and Passive Circuit Element	
	1.3	Representations of Basic Components.	
	1.4	Revisiting some Physical Quantities (Force, Energy, Power, Charge, Current, Potential, Electric Field, Magnetic Flux Density)	
2	**Introduction to Systems & Signals**		**3**
	2.1	Linear, Non-linear, Time-Invariant, Time-Variant System	
	2.2	Continuous Time and Discrete Time Signals, Periodic and Aperiodic signals	
	2.3	DC, AC, Sinusoidal Signal, Complex Signals, Square Wave, Triangular Wave, Concept of Duty Cycle.	
	2.4	Unit Step, Unit Ramp, Representing a signal using step function	
3	**Circuit Elements and Laws**		**4**
	3.1	Sources: Ideal voltage and current sources, Controlled sources	
	3.2	Resistors, Resistivity, Conductors, Conductivity, Mobility, Insulators, Semiconductor, Capacitor, Inductor	
	3.3	Ohm’s Law, KCL, KVL	
	3.4	Node and Loop Equations solving to find V and I, Cramer’s Rule, Gauss-Jordan Method	
	3.5	Current Division, Voltage Division, Series and Parallel Combinations, Star-to-Delta Conversion, Delta-to-Star Conversion, Wheatstone Bridge.	
4	**Network Theorems**		**3**
	4.1	Thevenin and Norton’s theorems	

	4.2	Source Transformation. Max Power transfer. Superposition Theorem.	
	4.3	Input and Output resistances. Finding V and I in Resistive Circuits with Controlled Sources.	
5	**OP-AMP**		**5**
	5.1	Idea of Amplifiers. OP Amps. Ideal OP Amps. Inverting and noninverting Amplifiers with Resistors.	
	5.2	Adders, Difference Amplifiers, Instrumentation Amplifiers	
	5.3	Concepts of common mode gain, differential gain, CMRR, Voltage saturation, Slew Rate, Offset Voltage	
	5.4	Designing controlled sources using Op-AMP, Analysing Linear Circuits with OP Amps.	
6	**Diode**		**3**
	6.1	n and p type semiconductor, Idea of p-n Junction	
	6.2	Semiconductor Diodes with Forward and Reverse Bias. Ideal Diode	
	6.3	Half-wave rectifiers, Full-wave rectifiers	
7	**Energy Storage Elements**		**2**
	7.1	Capacitors, Energy Storage in Capacitors, Series and Parallel Capacitors	
	7.2	Inductors, Energy Storage in Inductors, Series and Parallel Inductors	
	7.3	Op-Amp circuits with Capacitors and Inductors	
8	**Basic RL and RC Circuits**		**4**
	8.1	The source-free RL Circuit, Properties of exponential response	
	8.2	Source-free RC circuit, Concept of Time constant	
	8.3	Driven RL and RC circuits, Natural and Forced Response	
9	**Sinusoidal Steady-State Analysis**		**6**
	9.1	Characteristics of Sinusoids, Period, Frequency, Amplitude, Average and RMS values, Phase, Phase lag and lead	
	9.2	Forced Response to Sinusoidal Function	

	9.3	The Complex Forcing Function	
	9.4	The Phasor, Phasor Relationships for R, L, and C	
	9.5	Impedance, Admittance	
	9.6	Circuit laws and Network Theorems	
10	**Frequency Sensitive Circuits – Filters**		**4**
	10.1	Concept of Low Pass, High Pass, Band Pass and Band Stop Filters	
	10.2	Frequency Response of a Filter, Designing LPF, HPF, BPF and BSF using passive components	
	10.3	The concept of cut-off frequency, Bandwidth, Quality Factor	
	10.4	Active Filters, Designing 1^{st} and 2^{nd} order active filters using OP-AMP.	
11	**BJT**		**6**
	11.1	Operating regions of BJT, Applications of BJT	
	11.2	Biasing Circuits (Fixed Bias and Potential Divider)	
	11.3	Load-line, Q-point	
	11.4	Small signal operation and models	
	11.5	CE, CB, CC amplifiers focus on input resistance, output resistance and gain	